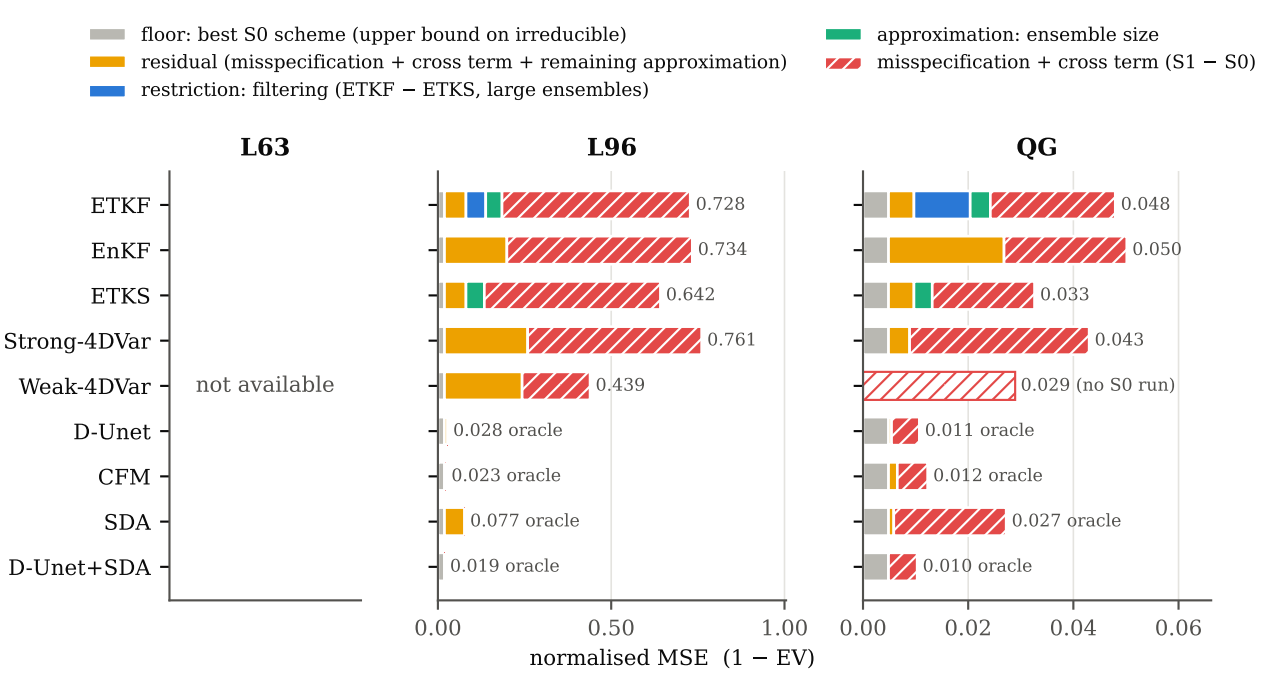
floor: best S0 scheme (upper bound on irreducible)
residual (misspecification + cross term + remaining approximation)
restriction: filtering (ETKF − ETKS, large ensembles)
approximation: ensemble size
misspecification + cross term (S1 − S0)
L63
L96
QG
ETKF
EnKF
ETKS
Strong-4DVar
Weak-4DVar
D-Unet
CFM
SDA
D-Unet+SDA
not available
0.728
0.734
0.642
0.761
0.439
0.028 oracle
0.023 oracle
0.077 oracle
0.019 oracle
0.048
0.050
0.033
0.043
0.029 (no S0 run)
0.011 oracle
0.012 oracle
0.027 oracle
0.010 oracle
0.00
0.50
1.00
0.00
0.02
0.04
0.06
normalised MSE (1 − EV)